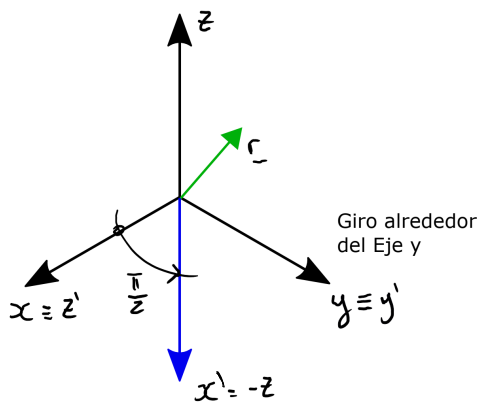
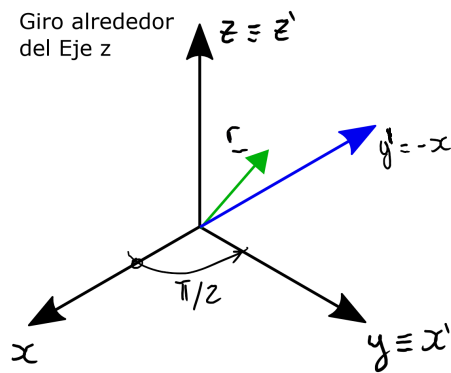




Rotación alrededor del Eje y



Rotación alrededor del Eje z

$$\underline{\beta}_1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\underline{\beta}_2 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\beta_{ij} = \underline{e}_i \cdot \underline{e}_j = \frac{1}{\|\underline{e}_i\|} \cdot \frac{1}{\|\underline{e}_j\|} \cdot \cos \theta = \cos \theta \quad 0 \leq \theta \leq \pi$$

$$\begin{matrix} \{x'\} & \{x\} \\ \uparrow & \uparrow \\ \underline{r}' & = \underline{\beta} \underline{r} \end{matrix}$$

$$\begin{bmatrix} r'_1 \\ r'_2 \\ r'_3 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$\underline{r} \rightarrow \{x'\} \rightarrow \begin{bmatrix} r'_1 \\ r'_2 \\ r'_3 \end{bmatrix} = \begin{bmatrix} -r_3 \\ r_2 \\ r_1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$\begin{bmatrix} r'_1 \\ r'_2 \\ r'_3 \end{bmatrix} = \begin{bmatrix} r_2 \\ -r_1 \\ r_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

- Primera transformación: $\underline{r} \xrightarrow{\beta_1} \underline{r}' \xrightarrow{\beta_2} \underline{r}''$

$$\left. \begin{matrix} \underline{r}' = \beta_1 \underline{r} \\ \underline{r}'' = \beta_2 \underline{r}' \end{matrix} \right\} \Rightarrow \underline{r}'' = \beta_2 \cdot \beta_1 \cdot \underline{r}$$

- Segunda transformación: $\underline{r} \xrightarrow{\beta_2} \underline{r}''' \xrightarrow{\beta_1} \underline{r}^{IV}$

$$\left. \begin{matrix} \underline{r}''' = \beta_2 \underline{r} \\ \underline{r}^{IV} = \beta_1 \underline{r}''' \end{matrix} \right\} \Rightarrow \underline{r}^{IV} = \beta_1 \cdot \beta_2 \cdot \underline{r}$$

Para que esta rotación sea conmutativa debería cumplirse que $\underline{r}'' = \underline{r}^{IV} \quad \forall \underline{r}$:

$$\underline{r}'' = \underline{r}^{IV} \quad \forall \underline{r} \Rightarrow \beta_2 \cdot \beta_1 \cdot \underline{r} = \beta_1 \cdot \beta_2 \cdot \underline{r} \quad \forall \underline{r} \Rightarrow \beta_2 \cdot \beta_1 = \beta_1 \cdot \beta_2$$

$$(\underline{\beta}_2 \cdot \underline{\beta}_1 - \underline{\beta}_1 \cdot \underline{\beta}_2) \underline{r} = \underline{0} \quad \forall \underline{r}$$

Por lo que se realiza las multiplicación matriciales:

$$\underline{\underline{B_{21}}} = \overbrace{\beta_2 \cdot \beta_1} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\underline{\underline{B_{12}}} = \overbrace{\beta_1 \cdot \beta_2} = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\underline{\underline{B_2}} \cdot \underline{\underline{B_1}} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$\underline{\underline{B_{21}}} \neq \underline{\underline{B_{12}}}$
 $\Downarrow$
Rotación no es conmutativa